

# Muhammad Jahanzaib Awan

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## PROFESSIONAL SUMMARY

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MSc Artificial Intelligence candidate (De Montfort University) specialising in deep learning, computer vision, and explainable AI for medical imaging. Hands-on experience building and benchmarking models across CNNs, Transformers, and sequence architectures in PyTorch, with a track record of distinction-level coursework (70–95%). Seeking a Machine Learning Engineer / Data Scientist graduate role to apply rigorous, evaluation-driven model development.

## EDUCATION

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**MSc Artificial Intelligence, De Montfort University, Leicester, UK** *Sep 2025 – Sep 2026 (expected)*

- Modules: Applied AI, Neural Systems, Natural Language Processing, Evolutionary Computing, Fuzzy Logic, Intelligent Mobile Robots, Mobile Robots, Research Methods, all assessed in the 70–95% range.

**BSc (Hons) Computer Science, Government College University Faisalabad, Pakistan** *2017 – 2021 · CGPA 3.09 / 4.0*

## EXPERIENCE

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**Bank Residential Support Worker, Platform Childcare Ltd, Leicester, UK** *Dec 2025 – Present*

- Provide specialist 1:1 support to children and young people with complex needs, including autism, learning disabilities, ADHD and physical disabilities, across residential, educational and community settings.
- Implement individual support plans and behaviour management strategies, adapting to each child's communication style and sensory needs, and support curriculum access through tailored learning activities.
- Maintain detailed daily records, incident reports and progress notes in line with safeguarding and regulatory requirements.
- Provide emotional support and de-escalation during challenging behaviour using trauma-informed approaches, and work in partnership with families as a trusted advocate for each child's needs.
- Supported children with profound and multiple learning disabilities (PMLD) to achieve educational and social goals.

**Founder and Sole Engineer, Automated Media Pipeline (independent project)** *Jun 2026 – Present*

Python · GitHub Actions · Cloudflare Workers · FFmpeg · React/Remotion · REST APIs

- Designed, built and operate an unattended content production system that has published autonomously twice daily for several months, from topic selection through scripted generation, media assembly, rendering and distribution to four platforms.
- Built a multi-provider LLM and text-to-speech layer with automatic failover between providers, so a quota exhaustion or billing lapse at one vendor degrades gracefully instead of halting production.
- Integrated and maintain OAuth-authenticated REST integrations across five external APIs, including token lifecycle management, scope auditing and expiry alerting after credential expiry caused repeated outages.
- Implemented automated quality gates that inspect each output before release, measuring transcription error rate against the source script, speech pace, loudness and frame-level visual completeness, and re-running generation until the output passes.
- Diagnosed a rendering fault that displayed blank frames over live audio by extracting and analysing frames from published output, traced it to a token-alignment defect, fixed it with regression tests and added the automated visual check that now prevents recurrence.

- Operate the system with scheduled external monitoring, committed run telemetry, automatic failure retry and alerting, plus a test suite run on every change.

## TECHNICAL PROJECTS

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### Anatomically-Gated Vision Transformers for Explainable Lung-Nodule Detection, MSc Final Project

PyTorch · MONAI · timm · Weights & Biases · 2026

- Designing a Vision Transformer that gates attention with anatomical lung masks to improve explainability of lung-nodule detection on chest radiographs (NIH ChestX-ray14, 112K+ images, 14 pathologies).
- Building a reproducible training pipeline with experiment tracking, transfer learning from pretrained backbones, and clinician-interpretable saliency outputs.

### Dialogue Generation: GRU vs LSTM vs GPT-2 (LoRA), Natural Language Processing

PyTorch · Hugging Face Transformers · PEFT/LoRA · 2026

- Built and compared three conversational models; a from-scratch Seq2Seq GRU achieved the best quality (perplexity 12.39, BLEU 3.35%, cosine similarity 0.40), outperforming a 355M-parameter GPT-2 by 2.8× on BLEU.
- Demonstrated that task-specific encoder-decoder design beat parameter-efficient fine-tuning of a larger pretrained model, with full evaluation across BLEU, perplexity and semantic similarity.

### UAV Aerial Object Classification, Applied AI

PyTorch · scikit-learn · 2026

- Built a 7-class image-classification dataset (8,903 cropped objects) from a 146-frame UAV sequence with severe class imbalance (car 42% vs bicycle 1%).
- Benchmarked ResNet18, MobileNetV2 and EfficientNet-B0 transfer learning against a HOG + SVM/Random-Forest classical baseline, reporting macro-F1 and per-class metrics.
- Engineered a leakage-aware, track-ID group split and quantified the accuracy inflation caused by naïve random splitting, the key methodological contribution.

### Mobile Robot Localization: AMCL vs EKF vs Hector SLAM, Intelligent Mobile Robots

ROS · Gazebo · TurtleBot3 · Python · 2026

- Deployed and compared three localization methods on a TurtleBot3 in simulation, logging Euclidean and orientation error against Gazebo ground truth across 1,100+ samples.
- EKF achieved 0.0047 m mean position error; analysed each method's trade-offs (odometry drift, scan-matching offset) to recommend AMCL for known-map deployment.

### Particle Swarm Optimization on Rastrigin, Evolutionary Computing

MATLAB · 2025

- Implemented PSO and tuned inertia/damping parameters, cutting convergence from 173 to 16 iterations, a 90.8% improvement, validated over multiple runs with summary statistics.

### Traffic-Signal Fuzzy Inference System, Fuzzy Logic

MATLAB Fuzzy Logic Toolbox · 2026

- Designed a Mamdani fuzzy controller for adaptive traffic-signal timing: membership-function design, operator/defuzzification comparison and system evaluation (assessed distinction).

**Additional projects:** Fashion-MNIST neural classifier (Neural Systems); PID wall-following controller in MATLAB + CoppeliaSim (Mobile Robots); empirical research proposal (Research Methods).

## TECHNICAL SKILLS

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**Languages:** Python, R, SQL, MATLAB

**ML / Deep Learning:** PyTorch, MONAI, timm, Hugging Face Transformers, PEFT/LoRA, scikit-learn

**Data & Analytics:** Tableau, data visualisation, exploratory data analysis, SQL querying

**Domains:** Computer Vision, Natural Language Processing, Medical Imaging, Explainable AI, Robotics, Data Analytics, Fuzzy Systems

**Tooling:** Git/GitHub, Weights & Biases, Google Colab, Jupyter, LaTeX, ROS, Gazebo, CoppeliaSim

*References available on request.*